UNIT I: Introduction to Data Warehousing (BCA Full Stack – V Semester)

Teaching Approach: Every topic is explained using:

- **Simple Definition**
- **Real-Life Analogy**
- **Technical Explanation**
- **Practical Example**
- **Industry Use Case**
- **Key Points**
- **Interview Question**

1. Introduction to Data Warehousing

What is Data?

Data is a collection of raw facts and figures.

Example

101, Rahul, Delhi, ₹45,000

102, Priya, Jaipur, ₹52,000

These are simply facts.

What is Information?

Information is processed data that has meaning.

Example

Average salary = ₹48,500

Highest salary = ₹52,000

Now data becomes useful.

What is a Data Warehouse?

Definition

A **Data Warehouse (DW)** is a centralized repository that stores large amounts of historical data collected from different sources for reporting and analysis.

Unlike normal databases that store current transactions, a Data Warehouse stores years of historical data.

Real-Life Analogy

School Library

Imagine a school.

- Every classroom has its own notebooks.
- Science teacher has science records.
- Maths teacher has maths records.
- Accounts office has fee records.
- Library has book records.

If the Principal wants to know

"Which students performed best over the last five years?"

He cannot visit every classroom individually.

Instead,

All records are copied into one central library.

That library is the **Data Warehouse**.

Science Records

    |

Math Records

    |

Fee Records

    |

Library Records



Technical Explanation

Different departments have different databases.

Sales Database

HR Database

Finance Database

Inventory Database

CRM Database

All these send their data into

Data Warehouse

Business Analysts generate reports from it.

Example

Amazon stores

- Customer Orders
- Payments
- Product Details
- Delivery Information
- Reviews

in different operational databases.

Every night,

all this information is transferred into a Data Warehouse.

Managers can now answer questions like

- Which product sold the most last year?
- Which city generated the highest revenue?
- Which customers purchase frequently?

Industry Use Cases

Amazon

Customer purchase analysis

Flipkart

Festival sales comparison

Netflix

Movie recommendation analysis

Hospitals

Patient treatment history

Universities

Student performance analysis

Key Features

✓ Centralized Storage

✓ Historical Data

✓ Multiple Data Sources

✓ Read Optimized

✓ Decision Making

2. Need for Data Warehousing

Why Do We Need It?

Companies generate millions of records daily.

These records are stored in different systems.

Without a Data Warehouse,

finding useful information becomes very difficult.

Real-Life Analogy

Imagine a student.

Notes are stored in

- WhatsApp
- Gmail
- Notebook
- Laptop
- Google Drive

Before exams,

finding everything becomes difficult.

Instead,

the student creates one folder containing everything.

That folder is the Data Warehouse.

Technical Explanation

Without Data Warehouse

Sales → Database A

HR → Database B

Finance → Database C

Need report?

Visit every database.

Time consuming.

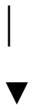
With Data Warehouse

Sales

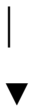
HR

Finance

Inventory



Data Warehouse



Single Report

Practical Example

A supermarket has

50 branches.

Manager wants

Total Milk Sales

Last 10 Years

Without Data Warehouse

Need to check

50 databases.

With Data Warehouse

One SQL Query.

Why Businesses Need It?

- Better Reporting
- Historical Analysis
- Business Decisions
- Data Integration
- Trend Prediction

Industry Use Case

Bank

Need to identify

Customers likely to take loans.

Historical transaction analysis is required.

3. Characteristics of Data Warehouse

According to the classic definition, a Data Warehouse has four major characteristics.

1. Subject-Oriented

Data is organized around subjects.

Examples

- Customers
- Products
- Sales
- Employees

Instead of organizing around applications.

Real-Life Analogy

Library books are arranged by

- Science
- Maths
- History

Not by publisher.

2. Integrated

Data comes from multiple sources.

Different formats become one standard format.

Example

Delhi

DELHI

New Delhi

Stored as

Delhi

Analogy

Different teachers write marks differently.

Office converts everything into one report card format.

3. Time Variant

Stores historical data.

Example

Sales

2022

2023

2024

2025

Operational database stores only current information.

Warehouse stores years of information.

Analogy

Photo Album

You don't keep only today's photo.

You keep memories of many years.

4. Non-Volatile

Once data enters the warehouse,

it is rarely updated or deleted.

Mostly

Read Only.

Analogy

University Result

After declaration,

students only view it.

No one edits it daily.

Interview Question

Explain Subject-Oriented and Non-Volatile with examples.

4. Operational Database vs Data Warehouse

Feature	**Operational Database (OLTP)**	**Data Warehouse (OLAP)**
Purpose	Daily operations	Business analysis
Data	Current	Historical
Users	Employees	Managers, Analysts
Operations	Insert, Update, Delete	Mostly Read
Speed	Fast transactions	Fast analytical queries
Size	Smaller	Very Large
Example	ATM transaction	Annual banking report

Real-Life Analogy

Restaurant

Kitchen

Receives orders every minute.

This is OLTP.

Owner

Checks monthly profits.

This is Data Warehouse.

Example

ATM

Withdraw ₹500

↓

Operational Database

Bank CEO

How many withdrawals happened in 2025?

↓

Data Warehouse

Industry Use Case

E-commerce

Customer places order

↓

OLTP

CEO checks yearly sales

↓

OLAP

5. Business Intelligence (BI)

Definition

Business Intelligence (BI) is a collection of technologies, processes, and tools that transform raw data into meaningful information for better business decisions.

BI uses the Data Warehouse as a major source of information.

Real-Life Analogy

A doctor receives reports like blood tests, X-rays, and scans. By analyzing all of them together, the doctor decides the best treatment.

Similarly, BI analyzes business data to support decisions.

Example

A retail company wants answers to questions such as:

- Which product generated the highest profit?
- Which city has the most customers?
- Which month has maximum sales?

BI dashboards provide these insights.

Industry Use Cases

- Sales performance dashboards
- Customer behavior analysis
- Financial reporting
- Marketing campaign analysis

Popular BI Tools

- Microsoft Power BI
- Tableau
- Qlik Sense
- SAP BusinessObjects
- IBM Cognos

Interview Question

What is Business Intelligence, and how does it use a Data Warehouse?

6. Decision Support Systems (DSS)

Definition

A **Decision Support System (DSS)** is a computer-based system that helps managers make informed decisions by analyzing data, models, and reports.

Real-Life Analogy

A GPS navigation app analyzes traffic, road closures, and distance to suggest the best route. The driver still makes the final decision.

Similarly, DSS provides recommendations, while managers make the final business decision.

Example

A bank uses DSS to evaluate a loan application by analyzing:

- Customer income
- Credit history
- Existing loans
- Repayment behavior

The system recommends whether the loan risk is low or high.

Industry Use Cases

- Loan approval
- Inventory planning
- Hospital resource allocation
- Airline ticket pricing

Interview Question

How is a Decision Support System different from an Operational Database?

7. Components of a Data Warehouse

A Data Warehouse consists of several key components:

1. **Data Sources** – Operational databases, ERP, CRM, files, APIs.
2. **ETL (Extract, Transform, Load)** – Collects, cleans, and loads data into the warehouse.
3. **Data Warehouse Storage** – Central repository for integrated historical data.
4. **Metadata** – Information about the data (data definitions, source, format).
5. **Data Marts** – Department-specific subsets (Sales, HR, Finance).
6. **BI and Reporting Tools** – Dashboards, reports, and analytics.

Real-Life Analogy

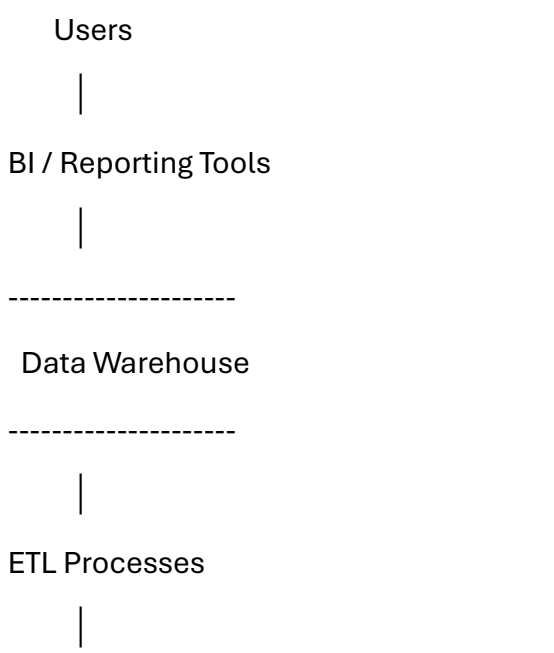
Think of preparing a meal:

- Ingredients = Data Sources
- Washing and cutting = ETL
- Refrigerator = Data Warehouse
- Labels on containers = Metadata

- Separate lunch boxes = Data Marts
- Serving the meal = BI Reports

8. Data Warehouse Architecture

Three-Tier Architecture



Bottom Tier

Data sources and ETL processes.

Middle Tier

Data Warehouse and Data Marts.

Top Tier

Reporting, dashboards, OLAP tools, and BI applications.

Industry Example

An online shopping company collects data from:

- Website
- Mobile App
- Payment Gateway
- Delivery System

ETL integrates the data into the warehouse, and managers view dashboards to monitor sales and customer behavior.

9. Data Warehouse Development Life Cycle

Typical stages include:

1. Requirement Gathering
2. Business Analysis
3. Data Source Identification
4. Data Modeling
5. ETL Design and Development
6. Data Warehouse Implementation
7. Testing
8. Deployment
9. Maintenance and Performance Tuning

Real-Life Analogy

Building a house:

- Plan the house
- Buy materials
- Build the structure
- Inspect quality
- Move in

- Maintain the house

A Data Warehouse follows a similar lifecycle.

10. Applications of Data Warehousing

Data Warehouses are used in many industries:

Industry	Application
Banking	Fraud detection, loan analysis
Healthcare	Patient history analysis
Education	Student performance analysis
Retail	Sales and inventory analysis
Manufacturing	Production planning
E-commerce	Customer behavior analysis
Telecommunications	Network usage analysis
Government	Population and tax analysis

11. Advantages of Data Warehousing

- Centralized data storage
- Historical data analysis
- Faster reporting
- Better decision-making
- Improved data quality
- Integration of multiple data sources
- Supports Business Intelligence
- Enables trend and predictive analysis
- Reduces report generation time

- Improves organizational efficiency

Real-Life Analogy

Instead of searching for documents in many cupboards, all documents are organized in one digital archive, making retrieval much faster.

12. Challenges of Data Warehousing

Some common challenges are:

- High implementation cost
- Large storage requirements
- Complex ETL processes
- Data quality issues
- Security and privacy concerns
- Long development time
- Performance optimization for huge datasets
- Continuous maintenance
- Integration of heterogeneous data sources

Example

If customer names are stored differently across systems ("Rahul Kumar", "R. Kumar", "Rahul K."), ETL must standardize them before loading into the warehouse.

13. Overview of Data Warehousing Tools

Tool	Description	Common Use
Microsoft SQL Server (SSIS)	ETL and data integration	Enterprise DW

Tool	Description	Common Use
Oracle Data Warehouse	Large-scale enterprise warehousing	Banking, Telecom
Snowflake	Cloud-native Data Warehouse	Modern analytics
Amazon Redshift	AWS cloud warehouse	Big Data analytics
Google BigQuery	Serverless cloud warehouse	Real-time analytics
Azure Synapse Analytics	Microsoft cloud analytics	Enterprise BI
Informatica PowerCenter	ETL tool	Data integration
Talend	Open-source ETL	Data migration
Pentaho	ETL and BI	Reporting and dashboards